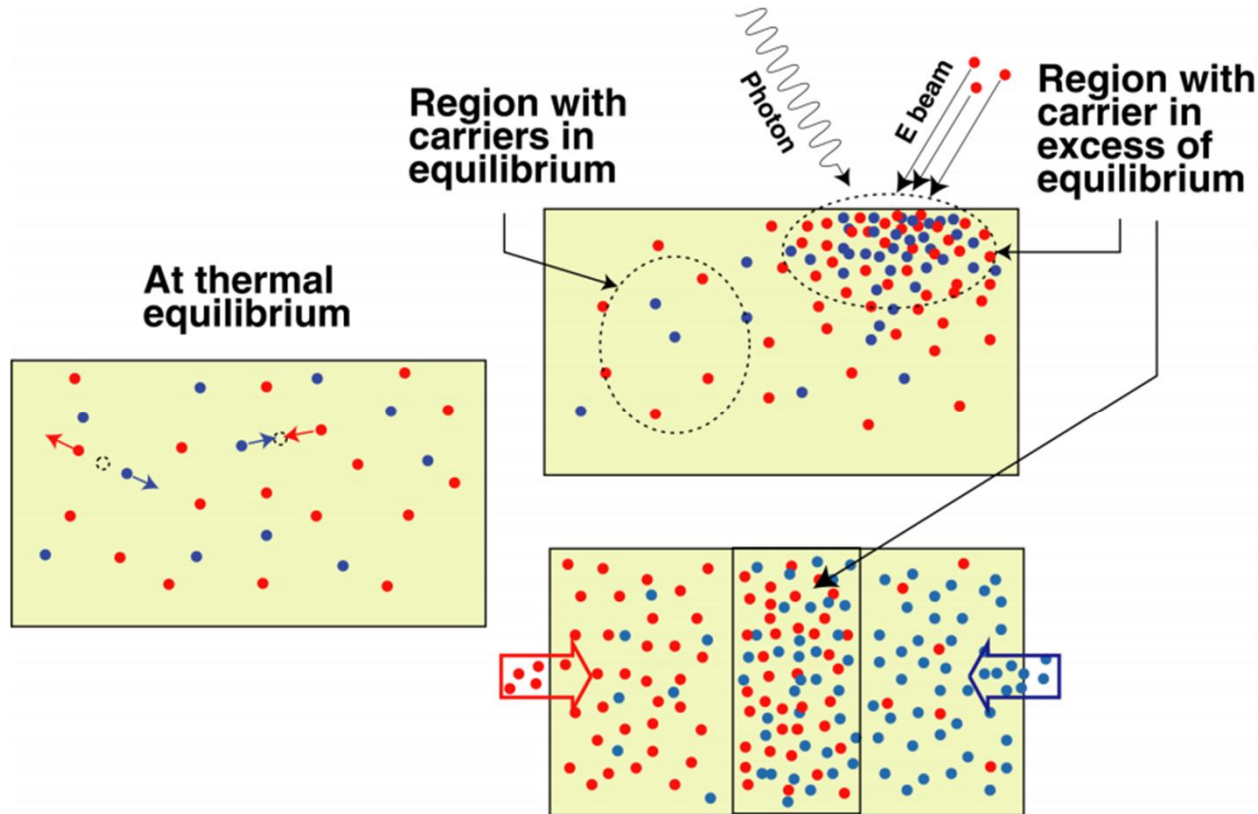


L6: Nonequilibrium Excess Carriers in Semiconductors



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Semiconductors under Nonequilibrium Conditions

- In Chapter 4 we talked about carrier distributions in semiconductors in thermal equilibrium conditions. In our discussion of current transport, we did not address nonequilibrium conditions but implicitly assumed that equilibrium was not significantly disturbed.
- Any deviation from thermal equilibrium will tend to change the electron and hole concentrations in a semiconductor. For example, a sudden increase in temperature will increase the rate at which electrons and holes are thermally generated so that their concentrations will change with time until new equilibrium values are reached. An external excitation, such as light, can also generate electrons and holes, creating a nonequilibrium condition.

Carrier Generation and Recombination in Semiconductors in Equilibrium

In thermal equilibrium, the electron and hole concentrations are independent of time. However, electrons are continually being thermally excited from the valence band into the conduction band by the random nature of the thermal process (**carrier generation**). At the same time, electrons moving randomly through the crystal in the conduction band can come in close proximity to holes and “fall” into the empty states in the valence band (**carrier recombination**).



In thermal equilibrium, the electron and hole concentrations are independent of time.

$$G_{n0} = G_{p0} = R_{n0} = R_{p0}$$

G_{n0} and G_{p0} are used to stand for the thermal generation rates of electrons and holes ($\text{cm}^{-3}\text{s}^{-1}$). R_{n0} and R_{p0} are used to stand for the recombination rates of electrons and holes ($\text{cm}^{-3}\text{s}^{-1}$). For direct band-to-band generation and recombination, the electrons and holes are generated and recombined in pairs.

$$G_{n0} = G_{p0}$$

$$R_{n0} = R_{p0}$$

Excess Carrier Generation and Recombination

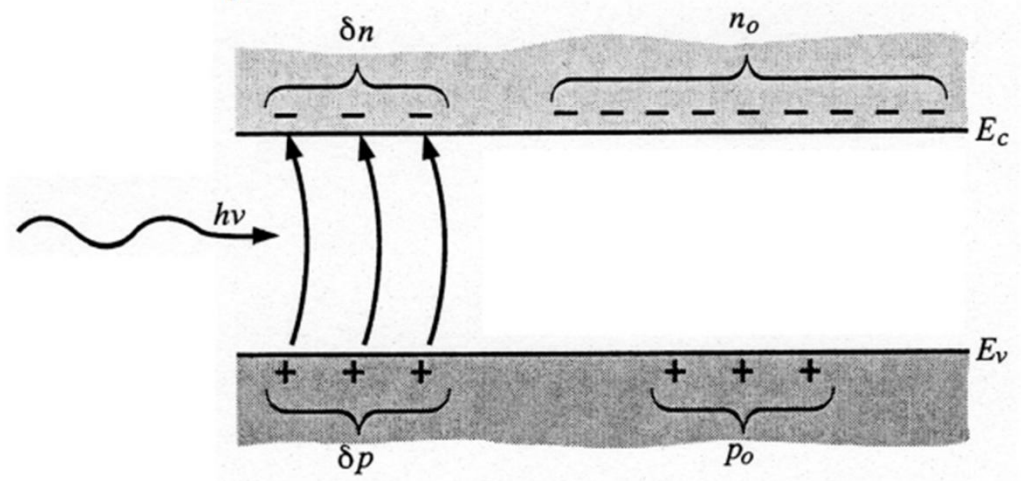
Electrons in the valence band may be excited into the conduction band by, for example, incident photons. When this happens, an electron-hole pair is generated. The additional electrons and holes created are called **excess electrons** and **excess holes**.

The excess electrons and holes are generated by an external force at particular rates, which are denoted by g_n' and g_p' .

For the direct band-to-band generation, the excess electrons and holes are created in pairs. We must have:

$$g_n' = g_p'$$

When excess electrons and holes are created, the concentrations of electrons in the conduction band and of holes in the valence band increase above their thermal equilibrium value.



Excess Carrier Generation and Recombination

$$n = n_0 + \delta n$$

$$p = p_0 + \delta p$$

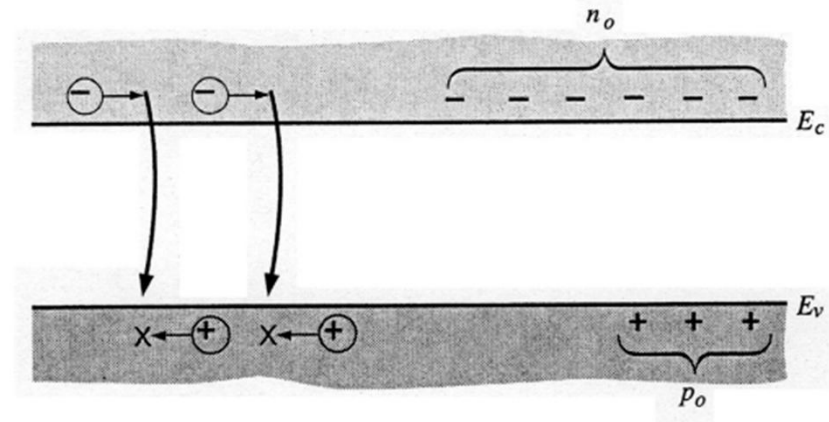
n_0 and p_0 are the thermal equilibrium concentrations, and δn and δp are the excess electron and hole concentrations.

The external force has perturbed the equilibrium condition so that the semiconductor is no longer in thermal equilibrium. It is noted that in a nonequilibrium condition,

$$np \neq n_0 p_0 = n_i^2$$

A steady-state generation of excess electrons and holes will not cause a buildup of the carrier concentrations, because of the process of excess electron-hole recombination. The recombination rates for excess electrons and holes are denoted by R_n' and R_p' .

$$R_n' = R_p'$$



Excess Carrier Generation and Recombination

In the direct band-to-band recombination, the recombination occurs spontaneously. The rate at which electrons recombine with holes must be proportional to the product of the electron and hole concentrations.

The net rate of change after an external perturbation in the electron concentration can be written as

$$\frac{dn(t)}{dt} = \underbrace{\alpha_i n_i^2}_{\text{generation}} - \underbrace{\alpha_r n(t)p(t)}_{\text{recombination}} = \alpha_r [n_i^2 - n(t)p(t)]$$

$$n(t) = n_0 + \delta n(t)$$

$$p(t) = p_0 + \delta p(t)$$

The first term is the thermal equilibrium generation rate. Because excess electrons and holes are created and recombined in pairs, we have that:

$$\delta n(t) = \delta p(t)$$

Excess Carrier Generation and Recombination

$$\begin{aligned}\frac{d[\delta n(t)]}{dt} &= \alpha_r [n_i^2 - (n_0 + \delta n(t))(p_0 + \delta p(t))] \\ &= -\alpha_r \delta n(t) [(n_0 + p_0) + \delta n(t)]\end{aligned}$$

The above equation can be solved if we impose the condition of **low-level injection**. Low-level injection means that the excess carrier concentration is much less than the thermal equilibrium majority carrier concentration.

If we consider a p-type material ($p_0 \gg n_0$) under low-level injection [$\delta n(t) \ll p_0$], then the equation becomes:

$$\frac{d[\delta n(t)]}{dt} = -\alpha_r p_0 \delta n(t)$$

The solution is an exponential decay from the initial excess minority carrier concentration. τ_{n0} is often referred to as the **excess minority carrier electron lifetime**.

$$\delta n(t) = \delta n(0)e^{-\alpha_r p_0 t} = \delta n(0)e^{-t/\tau_{n0}} \qquad \tau_{n0} = (\alpha_r p_0)^{-1}$$

Excess Carrier Generation and Recombination

The recombination rate of excess minority carrier electrons can be written as:

$$R'_n = \frac{-d[\delta n(t)]}{dt} = \alpha_r p_0 \delta n(t) = \frac{\delta n(t)}{\tau_{n0}}$$

For the direct band-to-band recombination, the excess majority carrier holes recombine at the same rate, so that:

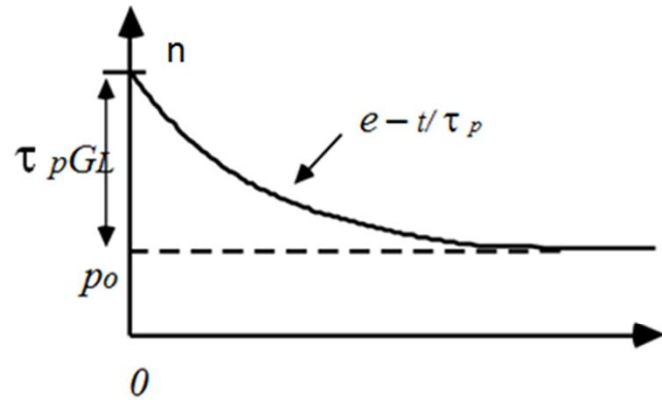
$$R'_n = R'_p = \frac{\delta n(t)}{\tau_{n0}}$$

In the case of an n-type material ($n_0 \gg p_0$) under low-level injection ($\delta n(t) \ll n_0$), the decay of excess minority carrier holes occurs with a time constant τ_{p0} . The recombination rate of the excess majority carrier electrons will be the same as that of the excess minority carrier holes, so we have:

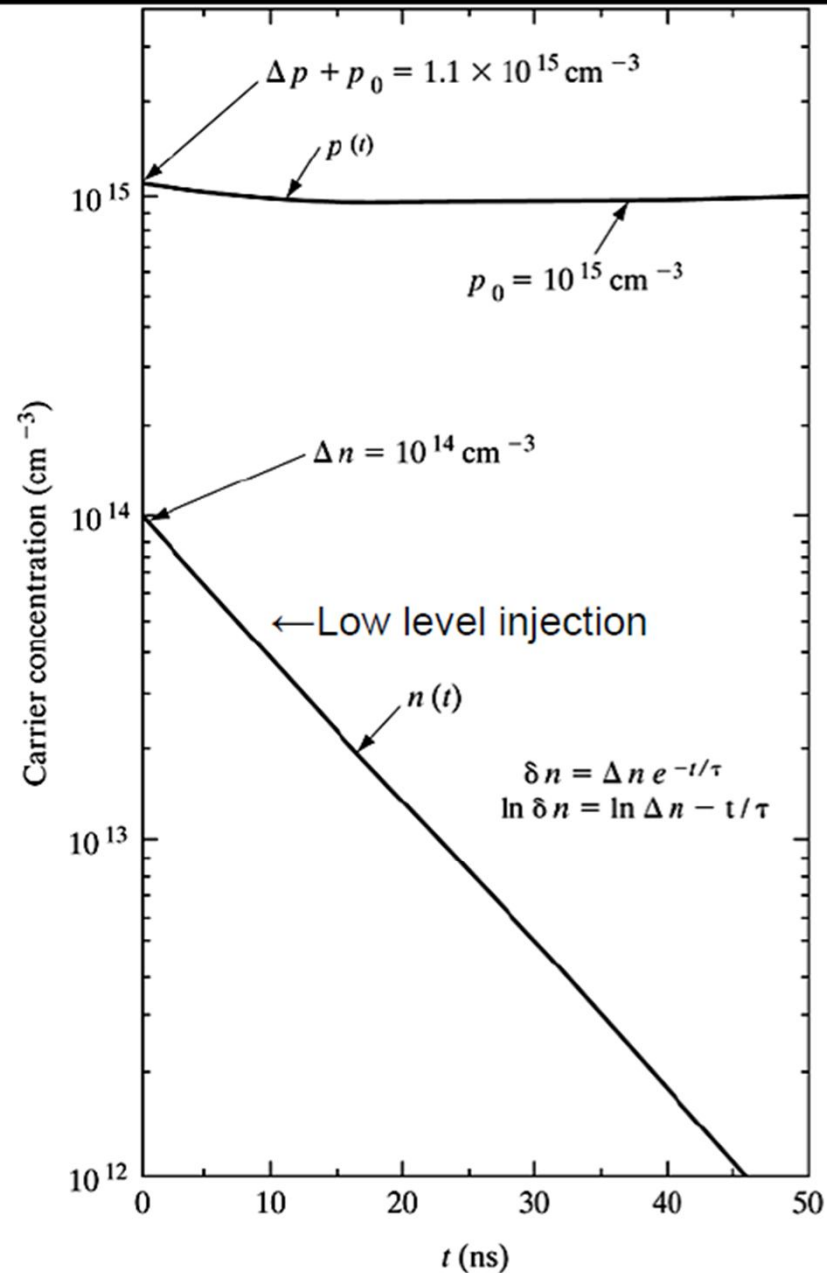
$$\tau_{p0} = \frac{1}{\alpha_r n_0}$$

$$R'_n = R'_p = \frac{\delta n(t)}{\tau_{p0}}$$

Excess Carrier Generation and Recombination



**Decay of
excess
carriers
through
recombination**



Indirect recombination

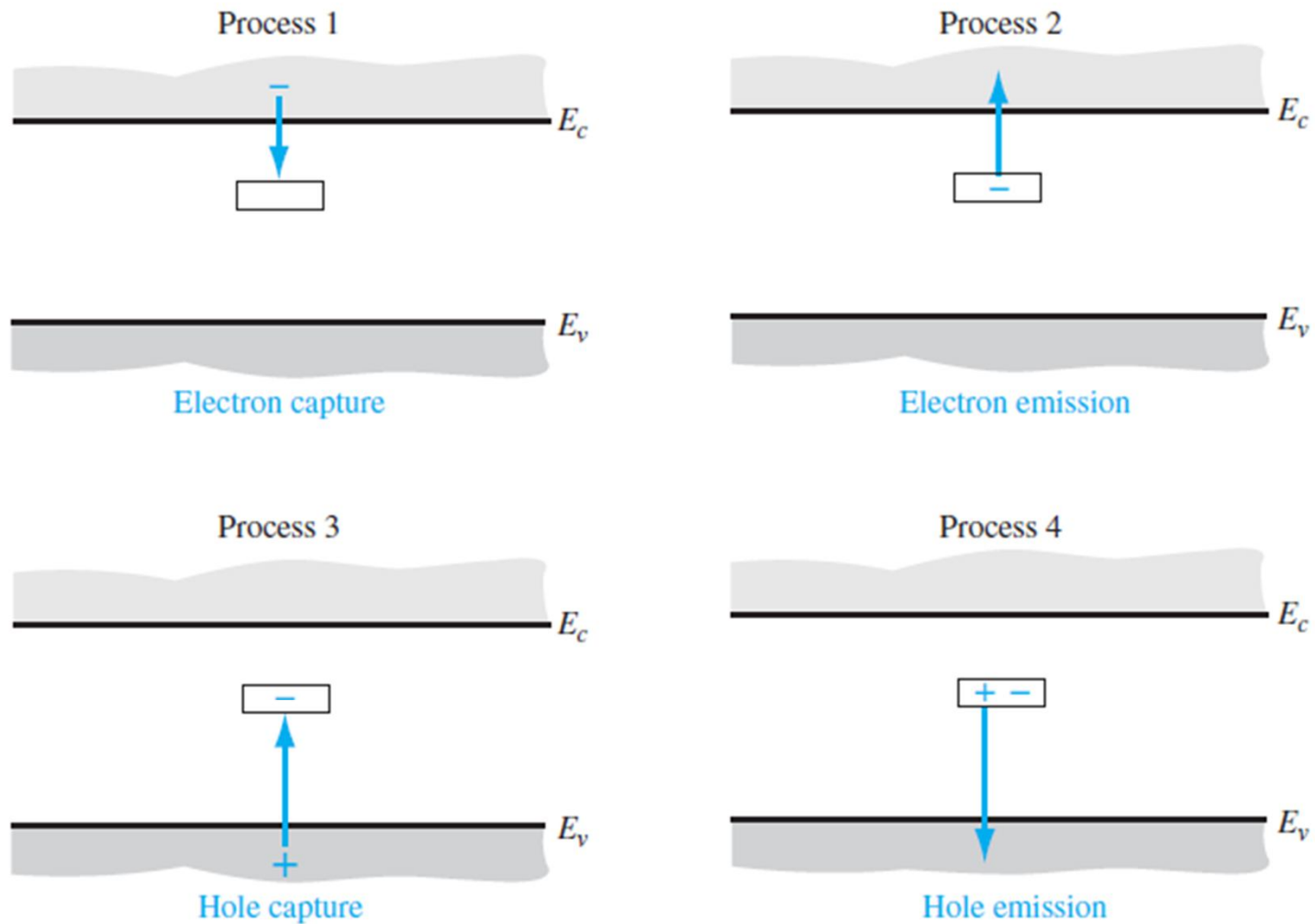
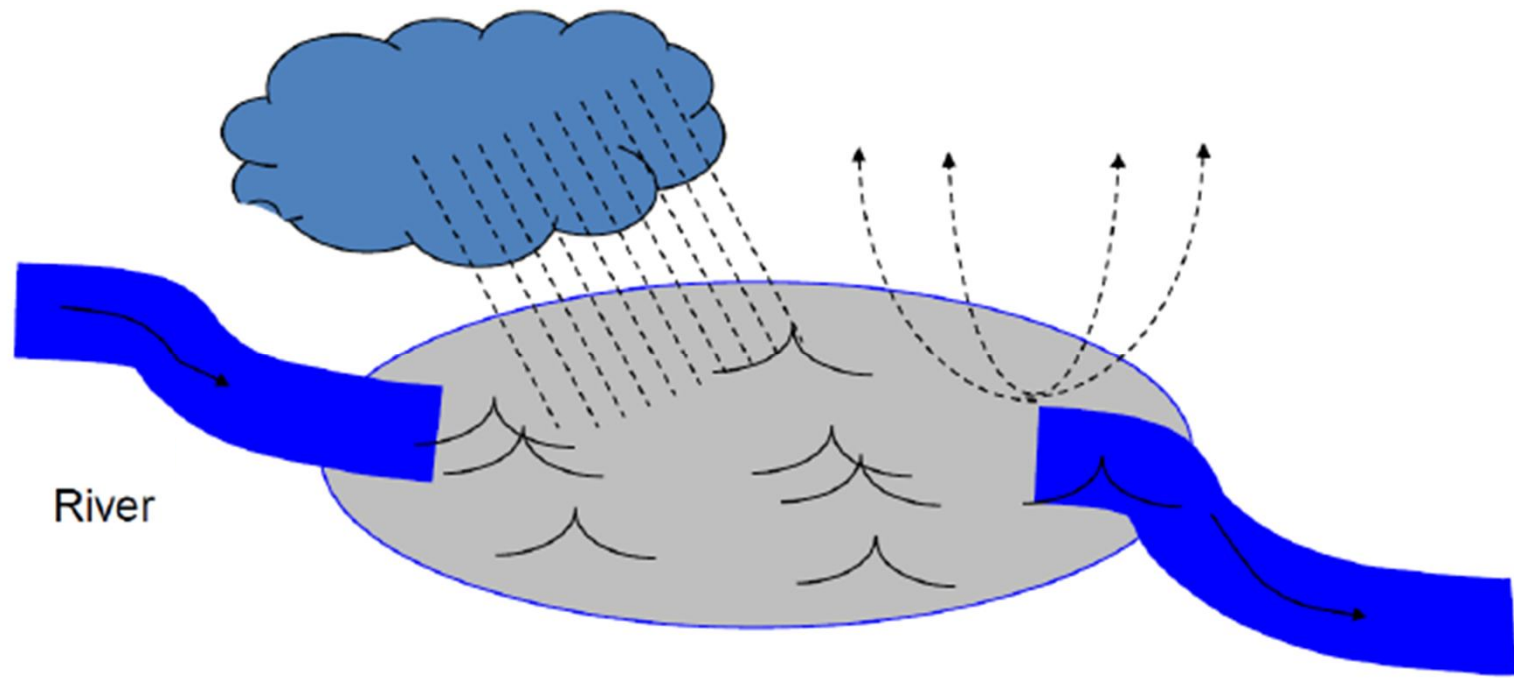


Figure 6.16 | The four basic trapping and emission processes for the case of an acceptor-type trap.

6.2.1 Continuity Equations

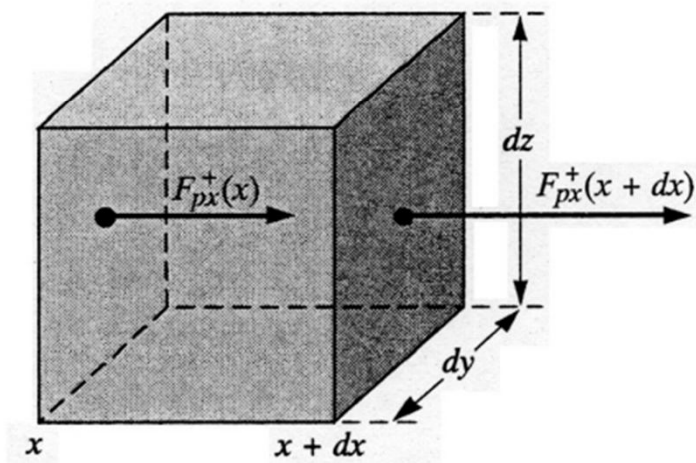


Rate of increase of
water level in lake = (in flow - outflow) + rain - evaporation

$$\frac{\partial n}{\partial t} = \frac{1}{q} \nabla \bullet J_n + g_N - r_N$$

Mathematical Analysis of Excess Carriers

The generation and recombination rates can be functions of time and space coordinates, and therefore the excess carrier concentration can also be a function of the same variables.



The left figure shows a differential volume element in which a one-dimensional hole-particle flux is entering the element at x and is leaving the element at $x + dx$. The parameter F_{px}^+ is the hole-particle flux, or flow (# of holes per cm^2 per s). We can write:

$$F_{px}^+(x + dx) = F_{px}^+(x) + \frac{\partial F_{px}^+}{\partial x} \cdot dx$$

The net increase in the number of holes per unit time within the element due to the x -component of hole flux is given by:

$$\frac{\partial p}{\partial t} dx dy dz = [F_{px}^+(x) - F_{px}^+(x + dx)] dy dz = -\frac{\partial F_{px}^+}{\partial x} dx dy dz$$

Mathematical Analysis of Excess Carriers

We can generalize the derivation to the three-dimensional case. However, we will limit ourselves to a one-dimensional analysis for simplicity.

The generation and recombination rate of holes will also affect the hole concentration in the differential volume. The net increase in the number of holes per unit time in the volume element is then given by:

$$\frac{\partial p}{\partial t} dx dy dz = -\frac{\partial F_{px}^+}{\partial x} dx dy dz + g_p dx dy dz - \frac{p}{\tau_{pt}} dx dy dz$$

The recombination rate for holes is given by p/τ_{pt} , where τ_{pt} includes the thermal equilibrium carrier lifetime and the excess carrier lifetime. The net increase in the hole concentration per unit time is obtained by dividing the differential volume:

Continuity equation for holes 

$$\frac{\partial p}{\partial t} = -\frac{\partial F_{px}^+}{\partial x} + g_p - \frac{p}{\tau_{pt}}$$

Mathematical Analysis of Excess Carriers

Similarly, the one-dimensional equation for electrons is:

$$\boxed{\frac{\partial n}{\partial t} = -\frac{\partial F_{nx}^-}{\partial x} + g_n - \frac{n}{\tau_{nt}}} \quad \leftarrow \text{Continuity equation for electrons}$$

In Chapter 4 we derived the hole and electron current densities in one dimension:

$$J_p = e\mu_p pE - eD_p \frac{\partial p}{\partial x}$$

$$J_n = e\mu_n nE + eD_n \frac{\partial n}{\partial x}$$

If we divide the hole current density by $(+e)$ and the electron current density by $(-e)$, we obtain each particle flux:

$$\frac{J_p}{+e} = F_p^+ = \mu_p pE - D_p \frac{\partial p}{\partial x}$$

$$\frac{J_n}{-e} = F_n^- = -\mu_n nE - D_n \frac{\partial n}{\partial x}$$

Mathematical Analysis of Excess Carriers

Substituting these two equations back into the continuity equations, we obtain:

$$\frac{\partial p}{\partial t} = -\mu_p \frac{\partial(pE)}{\partial x} + D_p \frac{\partial^2 p}{\partial x^2} + g_p - \frac{p}{\tau_{pt}}$$

$$\frac{\partial n}{\partial t} = +\mu_n \frac{\partial(nE)}{\partial x} + D_n \frac{\partial^2 n}{\partial x^2} + g_n - \frac{n}{\tau_{nt}}$$

We can expand the derivative of the product to obtain **the time-dependent diffusion equations** for holes and electrons, respectively. Both the hole and electron concentrations contain the excess carrier concentrations.

$$D_p \frac{\partial^2 p}{\partial x^2} - \mu_p \left(E \frac{\partial p}{\partial x} + p \frac{\partial E}{\partial x} \right) + g_p - \frac{p}{\tau_{pt}} = \frac{\partial p}{\partial t}$$
$$D_n \frac{\partial^2 n}{\partial x^2} + \mu_n \left(E \frac{\partial n}{\partial x} + n \frac{\partial E}{\partial x} \right) + g_n - \frac{n}{\tau_{nt}} = \frac{\partial n}{\partial t}$$

Mathematical Analysis of Excess Carriers

The hole and electron concentrations are functions of both the thermal equilibrium and the excess values. The thermal equilibrium concentrations, n_0 and p_0 , are independent of time. For the special case of a homogeneous semiconductor, n_0 and p_0 are also independent of the space coordinates. Then **the time-dependent diffusion equations** become:

$$D_p \frac{\partial^2(\delta p)}{\partial x^2} - \mu_p \left(E \frac{\partial(\delta p)}{\partial x} + p \frac{\partial E}{\partial x} \right) + g_p - \frac{p}{\tau_{pt}} = \frac{\partial(\delta p)}{\partial t}$$
$$D_n \frac{\partial^2(\delta n)}{\partial x^2} + \mu_n \left(E \frac{\partial(\delta n)}{\partial x} + n \frac{\partial E}{\partial x} \right) + g_n - \frac{n}{\tau_{nt}} = \frac{\partial(\delta n)}{\partial t}$$

6.3 | AMBIPOLAR TRANSPORT

During the derivation of the time-dependent diffusion equations, the electric field was assumed to be an applied electric field. If a pulse of excess electrons and a pulse of excess holes are created at a particular point in a semiconductor with an applied electric field, the excess holes and electrons will tend to drift in opposite directions. However, because electrons and holes are charged particles, any separation will induce an internal electric field between the two sets of particles. This internal electric field will create a force attracting the electrons and holes back toward each other. The electric field is then composed of the externally applied field plus the induced internal field:

$$E = E_{\text{app}} + E_{\text{int}}$$

Since the internal electric field creates a force attracting the electrons and holes, this field will hold the pulses of the excess electrons and excess holes together. The negatively charged electrons and positively charged holes will then drift or diffuse together with a single effective mobility or diffusion coefficient. This phenomenon is called **ambipolar diffusion** or **ambipolar transport**.

Derivation of the Ambipolar Transport Equation

To make the time-dependent diffusion equations more tractable, we need to make some approximations:

(1) Only a relatively small internal electric field is sufficient to keep the excess electrons and holes drifting and diffusing together. Hence:

$$|E_{\text{int}}| \ll |E_{\text{app}}|$$

(2) $\partial E / \partial x$ is non-negligible, but we are going to eliminate this term.

(3) The condition of charge neutrality will be imposed: the excess electron concentration is just balanced by an equal excess hole concentration at any point in space and time. Therefore:

$$\delta n \cong \delta p$$

(4) We can define:

$$g_n = g_p = g$$

$$R_n = \frac{n}{\tau_{nt}} = R_p = \frac{p}{\tau_{pt}} = R$$

The lifetimes here include the thermal equilibrium carrier lifetimes and the excess carrier lifetimes.

Derivation of the Ambipolar Transport Equation

We can rewrite the time-dependent diffusion equations on the basis of the approximations:

$$D_p \frac{\partial^2(\delta n)}{\partial x^2} - \mu_p \left(E \frac{\partial(\delta n)}{\partial x} + p \frac{\partial E}{\partial x} \right) + g - R = \frac{\partial(\delta n)}{\partial t}$$

$$D_n \frac{\partial^2(\delta n)}{\partial x^2} + \mu_n \left(E \frac{\partial(\delta n)}{\partial x} + n \frac{\partial E}{\partial x} \right) + g - R = \frac{\partial(\delta n)}{\partial t}$$

We multiply the first equation by $\mu_n n$, the second equation by $\mu_p p$, and add the two equations. The result is:

$$\begin{aligned} & (\mu_n n D_p + \mu_p p D_n) \frac{\partial^2(\delta n)}{\partial x^2} + (\mu_n \mu_p)(p - n) E \frac{\partial(\delta n)}{\partial x} \\ & + (\mu_n n + \mu_p p)(g - R) = (\mu_n n + \mu_p p) \frac{\partial(\delta n)}{\partial t} \end{aligned}$$

Derivation of the Ambipolar Transport Equation

If we divide the equation by the term $(\mu_n n + \mu_p p)$ and let:

$$D' = \frac{\mu_n n D_p + \mu_p p D_n}{\mu_n n + \mu_p p} = \frac{D_n D_p (n + p)}{D_n n + D_p p} \qquad \mu' = \frac{\mu_n \mu_p (p - n)}{\mu_n n + \mu_p p}$$

We obtain:

$$D' \frac{\partial^2(\delta n)}{\partial x^2} + \mu' E \frac{\partial(\delta n)}{\partial x} + g - R = \frac{\partial(\delta n)}{\partial t}$$

This equation is called the **ambipolar transport equation** and describes the behavior of the excess electrons and holes in time and space. The parameter D' is called the **ambipolar diffusion coefficient** and μ' is called the **ambipolar mobility**.

Since the ambipolar diffusion coefficient and the ambipolar mobility are functions of the electron and hole concentrations, which contain the excess carrier concentrations, the ambipolar transport equation is a nonlinear differential equation.

The Case of Extrinsic Doping and Low Injection

The ambipolar transport equation can be simplified and linearized by considering an extrinsic semiconductor and by considering low-level injection. The ambipolar diffusion coefficient can be rewritten as:

$$D' = \frac{D_n D_p (n + p)}{D_n n + D_p p} = \frac{D_n D_p [(n_0 + \delta n) + (p_0 + \delta n)]}{D_n (n_0 + \delta n) + D_p (p_0 + \delta n)}$$

If we consider a p-type semiconductor under the condition of low-level injection and assume that D_n and D_p , μ_n and μ_p are on the same order of magnitude, respectively, then:

$$n_0 \ll p_0 \qquad \delta n \ll p_0$$

$$D' \approx \frac{D_n D_p p_0}{D_p (p_0)} = D_n$$

$$\mu' = \frac{\mu_n \mu_p (p - n)}{\mu_n n + \mu_p p} = \frac{\mu_n \mu_p [(p_0 + \delta n) - (n_0 + \delta n)]}{\mu_n (n_0 + \delta n) + \mu_p (p_0 + \delta n)} \approx \frac{\mu_n \mu_p p_0}{\mu_p p_0} = \mu_n$$

The Case of Extrinsic Doping and Low Injection

If we consider a n-type semiconductor under the condition of low-level injection, we have:

$$p_0 \ll n_0$$

$$\delta n \ll n_0$$

$$D' \approx D_p$$

$$\mu' \approx -\mu_p$$

It is important to note that for an extrinsic semiconductor under low-level injection, the ambipolar diffusion coefficient and the ambipolar mobility reduce to the minority carrier parameter values, which are constants. The ambipolar transport equation therefore reduces to a linear differential equation with constant coefficients.

The Case of Extrinsic Doping and Low Injection

The remaining terms we need to consider in the ambipolar transport equation are the generation rate and the recombination rate. Recall that the electron and hole recombination rates are equal and given by:

$$R_n = \frac{n}{\tau_{nt}} = R_p = \frac{p}{\tau_{pt}} = R$$

τ_{nt} and τ_{pt} are the mean electron and hole lifetimes, respectively. If we consider their inverses, $1/\tau_{nt}$ is the probability per unit time that an electron will encounter a hole and recombine, and $1/\tau_{pt}$ is the probability per unit time that a hole will encounter an electron and recombine.

If we consider again an extrinsic p-type semiconductor, the concentration of majority carrier holes will be essentially constant. Then, the probability per unit time of a minority carrier electron encountering a majority carrier hole will be essentially constant. Hence $\tau_{nt} = \tau_{n0}$, the minority carrier electron lifetime, will remain a constant for the extrinsic p-type semiconductor under low-level injection.

The Case of Extrinsic Doping and Low Injection

For electrons:

$$g - R = g_n - R_n = (G_{n0} + g'_n) - (R_{n0} + R'_n)$$

G_{n0} and g'_n are the thermal equilibrium electron and excess electron generation rates, respectively. R_{n0} and R'_n are the thermal equilibrium electron and excess electron recombination rates, respectively. For thermal equilibrium, we have:

$$G_{n0} = R_{n0}$$

$$g - R = g'_n - R'_n = g'_n - \frac{\delta n}{\tau_{n0}}$$

For holes:

$$G_{p0} = R_{p0}$$

$$g - R = g_p - R_p = (G_{p0} + g'_p) - (R_{p0} + R'_p) = g'_p - R'_p = g'_p - \frac{\delta p}{\tau_{p0}}$$

τ_{n0} and τ_{p0} are the excess minority carrier electron and hole lifetimes, respectively.

The Case of Extrinsic Doping and Low Injection

The generation rate for excess electrons must equal to the generation rate for excess holes. Therefore, $g_n' = g_p' = g'$. In addition, the minority carrier lifetime is essentially a constant for low-level injection. Then the term $g - R$ in the ambipolar transport equation can be written in terms of the minority carrier parameters.

The ambipolar transport equation for an extrinsic p-type semiconductor under low injection then becomes:

$$D_n \frac{\partial^2(\delta n)}{\partial x^2} + \mu_n E \frac{\partial(\delta n)}{\partial x} + g' - \frac{\delta n}{\tau_{n0}} = \frac{\partial(\delta n)}{\partial t}$$

Similarly, for an extrinsic n-type semiconductor under low injection, the ambipolar transport equation becomes:

$$D_p \frac{\partial^2(\delta p)}{\partial x^2} - \mu_p E \frac{\partial(\delta p)}{\partial x} + g' - \frac{\delta p}{\tau_{p0}} = \frac{\partial(\delta p)}{\partial t}$$

The Importance of Ambipolar Transport Equation

- It is extremely important to note that the transport and recombination parameters in the ambipolar transport equations are **those of the minority carriers**.
- The ambipolar transport equations describe the drift, diffusion, and recombination of excess minority carriers as a function of spatial coordinates and time.
- The condition of charge neutrality implies that the excess majority carriers diffuse and drift together with the excess minority carriers. **The behavior of the excess majority carriers is determined by the minority carrier parameters.**
- This ambipolar phenomenon is extremely important in semiconductor physics, and is the basis for describing the characteristics and behavior of semiconductor devices.

Applications of the Ambipolar Transport Equations

Table 6.2 | Common ambipolar transport equation simplifications

Specification	Effect
Steady state	$\frac{\partial(\delta n)}{\partial t} = 0, \quad \frac{\partial(\delta p)}{\partial t} = 0$
Uniform distribution of excess carriers (uniform generation rate)	$D_n \frac{\partial^2(\delta n)}{\partial x^2} = 0, \quad D_p \frac{\partial^2(\delta n)}{\partial x^2} = 0$
Zero electric field	$E \frac{\partial(\delta n)}{\partial x} = 0, \quad E \frac{\partial(\delta p)}{\partial x} = 0$
No excess carrier generation	$g' = 0$
No excess carrier recombination (infinite lifetime)	$\frac{\delta n}{\tau_{n0}} = 0, \quad \frac{\delta p}{\tau_{p0}} = 0$

Applications of the Ambipolar Transport Equations

| Example 6.1

Homogenous n - type semiconductor

zero applied electrical field

under low - injection condition

At $t = 0$, uniform concentration of excess carriers

$g' = 0$ for $t > 0$

$\Rightarrow$ Calculate the excess carrier concentration as a function of time

Solution:

$$D_p \frac{\partial^2(\delta p)}{\partial x^2} - \cancel{\mu_p E \frac{\partial(\delta p)}{\partial x}} + \cancel{g'} - \frac{\delta p}{\tau_{po}} = \frac{\partial(\delta p)}{\partial t}$$

(Note: In the original image, arrows point from E=0 and g'=0 to the crossed-out terms in the equation above.)

$$D_p \frac{\partial^2(\delta p)}{\partial x^2} - \frac{\delta p}{\tau_{po}} = \frac{\partial(\delta p)}{\partial t}$$

Uniform excess carriers

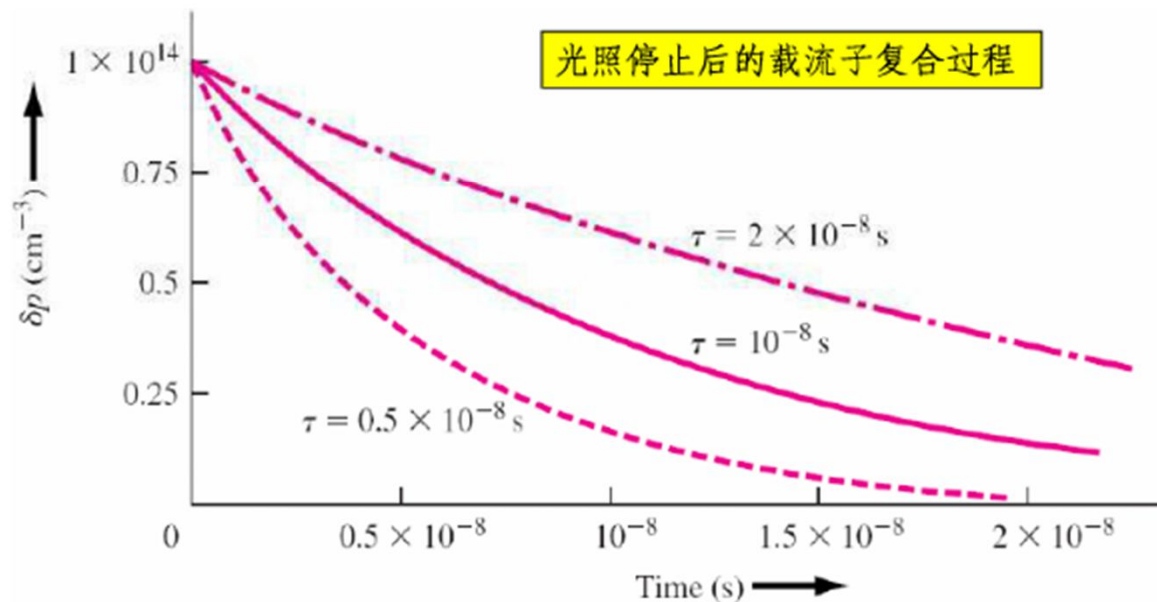
Applications of the Ambipolar Transport Equations

$$\frac{\partial(\delta p)}{\partial t} = -\frac{\delta p}{\tau_{po}}$$

$$\delta p(t) = \delta p(0)e^{-t/\tau_{po}}$$

From the **charge neutrality condition**, the excess majority electron concentration is given by

$$\delta n(t) = \delta n(0)e^{-t/\tau_{po}}$$



Applications of the Ambipolar Transport Equations

Example 6.2

Homogenous n - type semiconductor

zero applied electrical field

under low - injection condition

At $t > 0$, uniform generation of excess carriers

⇒ Calculate the excess carrier concentration as a function of time

Solution: Uniform generation

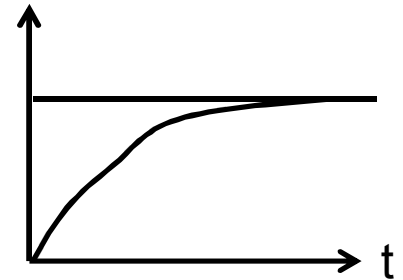
$$D_p \frac{\partial^2(\delta p)}{\partial x^2} - \mu_p E \frac{\partial(\delta p)}{\partial x} + g' - \frac{\delta p}{\tau_{po}} = \frac{\partial(\delta p)}{\partial t}$$

$$g' - \frac{\delta p}{\tau_{po}} = \frac{\partial(\delta p)}{\partial t} \Rightarrow \frac{\partial(\delta p)}{\partial t} + \frac{\delta p}{\tau_{po}} - g' = 0$$

$$\delta p(t) = \tau_{po} g' (1 - e^{-t/\tau_{po}})$$

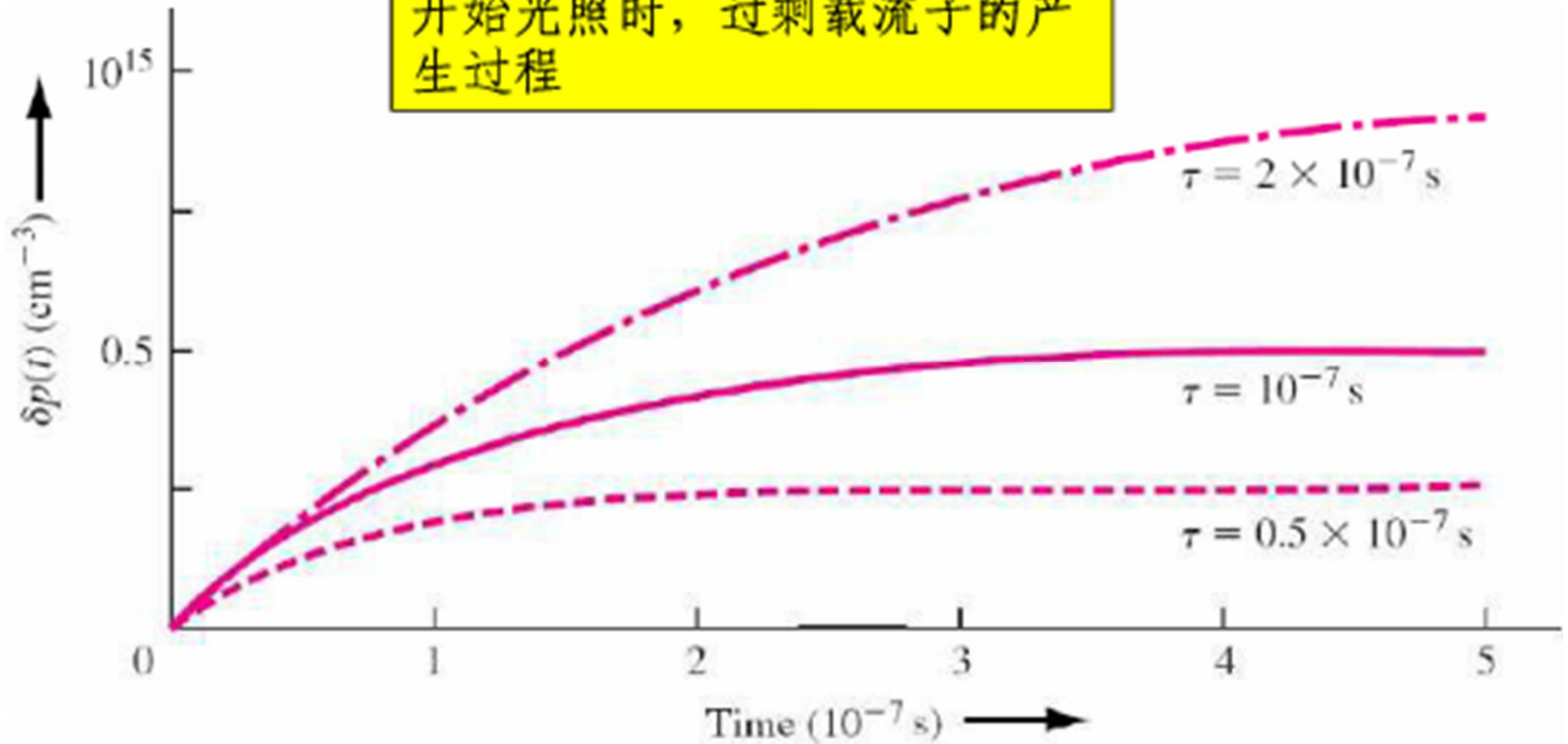
At steady state,

$$\frac{\partial \delta p(t)}{\partial t} = 0, \quad \delta p(t = \infty) = \tau_{po} g'$$



Applications of the Ambipolar Transport Equations

开始光照时，过剩载流子的产生过程



Applications of the Ambipolar Transport Equations

Example: assume that a finite number of electron-hole pairs is generated instantaneously at time $t = 0$ and at $x = 0$, but assume $g' = 0$ for $t > 0$ for an n-type semiconductor. A constant applied electric field E_0 is applied in the $+x$ direction to this semiconductor. Calculate the excess carrier concentration as a function of x and t .

Solution: the one-dimensional ambipolar transport equation for the minority carrier holes can be written as:

$$D_p \frac{\partial^2(\delta p)}{\partial x^2} - \mu_p E_0 \frac{\partial(\delta p)}{\partial x} - \frac{\delta p}{\tau_{p0}} = \frac{\partial(\delta p)}{\partial t}$$

The solution to this partial differential equation is of the form:

$$\delta p(x, t) = p'(x, t) e^{-t/\tau_{p0}}$$

Applications of the Ambipolar Transport Equations

We are left by substitution with the partial differential equation:

$$D_p \frac{\partial^2 p'(x,t)}{\partial x^2} - \mu_p E_0 \frac{\partial p'(x,t)}{\partial x} = \frac{\partial p'(x,t)}{\partial t}$$

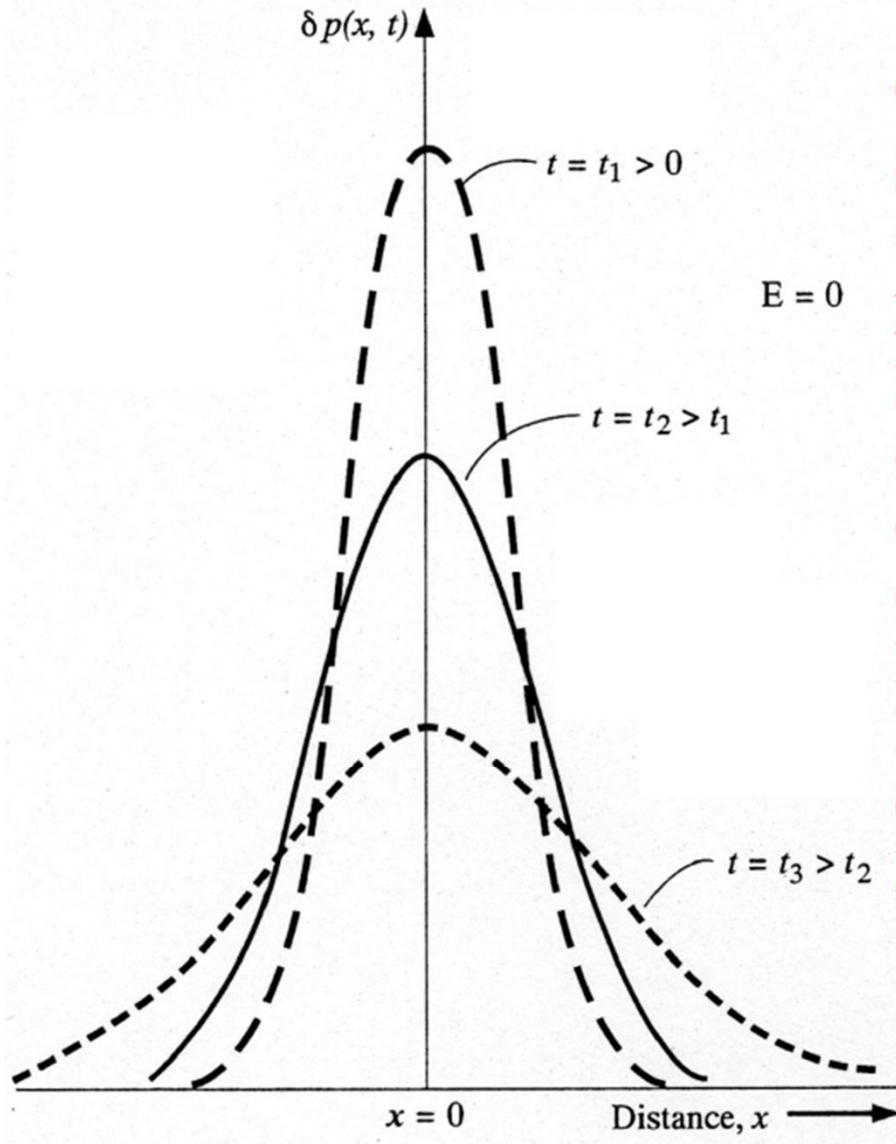
The solution obtained using Laplace transform techniques is:

$$p'(x,t) = \frac{1}{(4\pi D_p t)^{1/2}} \exp\left[\frac{-(x - \mu_p E_0 t)^2}{4D_p t}\right]$$

The total solution for the excess minority carrier hole concentration is:

$$\delta p(x,t) = \frac{e^{-t/\tau_{p0}}}{(4\pi D_p t)^{1/2}} \exp\left[\frac{-(x - \mu_p E_0 t)^2}{4D_p t}\right]$$

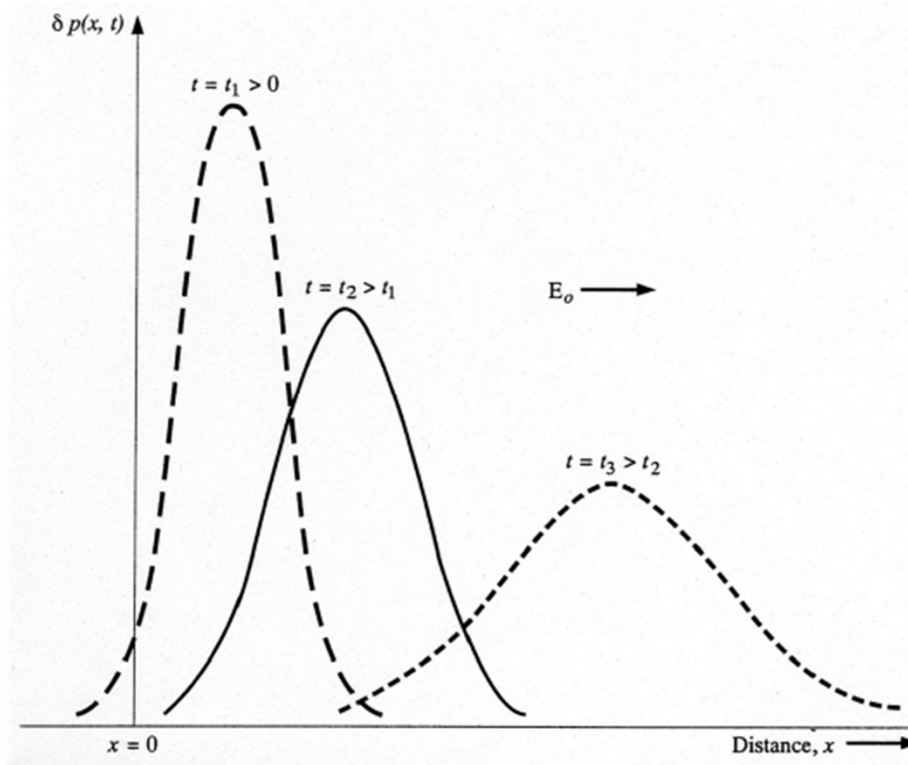
Applications of the Ambipolar Transport Equations



The left plot shows the case when the applied electric field is zero.

For $t > 0$, the excess minority carrier holes diffuse in both the $+x$ and $-x$ directions. During this time, the excess majority carrier electrons diffuse at exactly the same rate as the holes. As time proceeds, the excess holes recombine with the excess electrons so that at $t = \infty$ the excess hole concentration is zero.

Applications of the Ambipolar Transport Equations



The upper plot shows the case when the applied electric field is not zero. In this case, the pulse of excess minority carrier holes is drifting in the $+x$ direction. An important point is that the excess-electron pulse is moving in the same direction as the applied electric field, even though the electrons have a negative charge.

The excess majority carrier electrons are being pulled along by the excess minority carrier holes.

6.4 | QUASI-FERMI ENERGY LEVELS

The thermal-equilibrium electron and hole concentrations are functions of the Fermi energy level. We can write

$$n_o = N_C \exp\left[\frac{-(E_C - E_F)}{kT}\right] = n_i \exp\left[\frac{E_F - E_{Fi}}{kT}\right]$$

$$p_o = N_v \exp\left[\frac{-(E_F - E_v)}{kT}\right] = n_i \exp\left[\frac{-(E_F - E_{Fi})}{kT}\right]$$

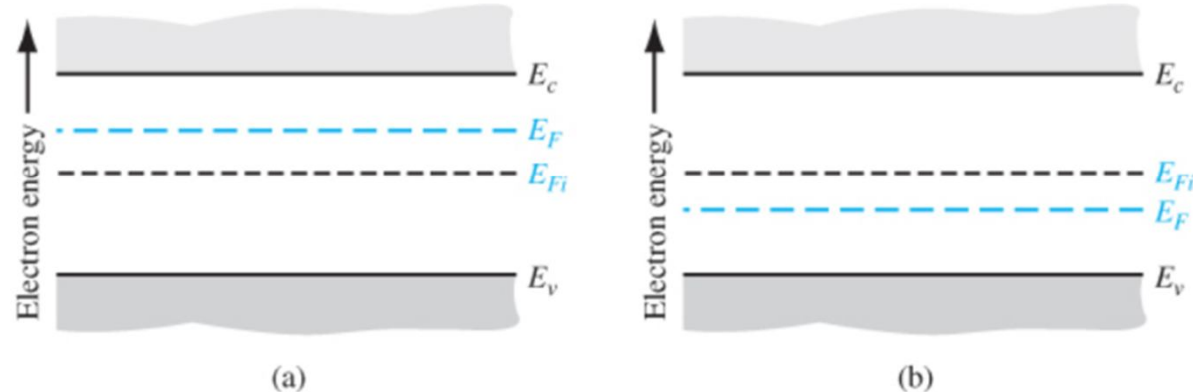


Figure 6.14 | Thermal-equilibrium energy-band diagrams for (a) n-type semiconductor and (b) p-type semiconductor.

QUASI-FERMI ENERGY LEVELS

If excess carriers are created in a semiconductor, we are no longer in thermal equilibrium and the Fermi energy is strictly no longer defined. If δn and δp are excess electron and hole concentrations respectively, we write

$$n_o + \delta n = n_i \exp \left[\frac{E_{Fn} - E_{Fi}}{kT} \right]$$

$$p_o + \delta p = n_i \exp \left[\frac{-(E_{Fp} - E_{Fi})}{kT} \right]$$

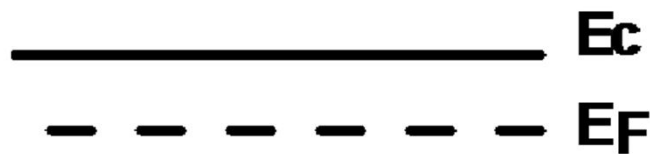
QUASI-FERMI ENERGY LEVELS

where E_{Fn} = electron quasi Fermi level

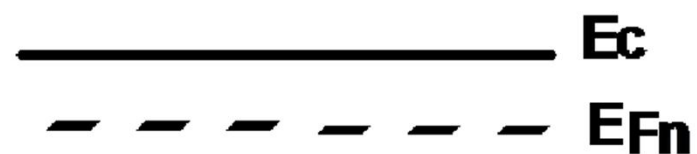
E_{Fp} = hole quasi Fermi level

Example: Generation through *low-level* light irradiation in n-type silicon.

$$n \approx n_{no} \quad \text{but} \quad p \gg p_{no}$$



Thermal
Equilibrium



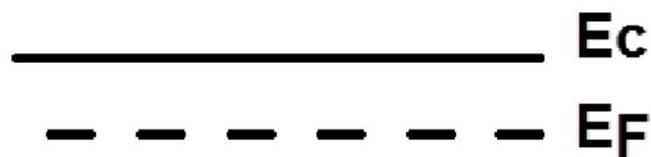
Low level
injection

QUASI-FERMI ENERGY LEVELS

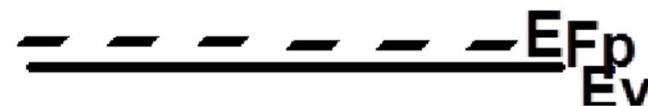
In general, $E_{Fn} \neq E_{Fp}$.

E_{Fn} and E_{Fp} are mathematical tools; their values are chosen so that the correct **carrier concentrations** are given in **non-equilibrium** situations. Note that the quasi-Fermi level of the majority carriers is essentially unchanged under low-level injection.

Under **high-level injection**, the quasi-Fermi levels **for both** carriers will differ from the Fermi level.



Thermal
Equilibrium



High level
Injection

QUASI-FERMI ENERGY LEVELS

Question: Consider an n-type semiconductor at $T=300\text{K}$ with carrier concentration of $n_0=10^{15}\text{ cm}^{-3}$, $n_i=10^{10}\text{ cm}^{-3}$, and $p_0=10^5\text{ cm}^{-3}$. In nonequilibrium, assume that the excess carrier concentrations are $\delta n=\delta p= 10^{13}\text{ cm}^{-3}$

The Fermi level for thermal equilibrium can be determined

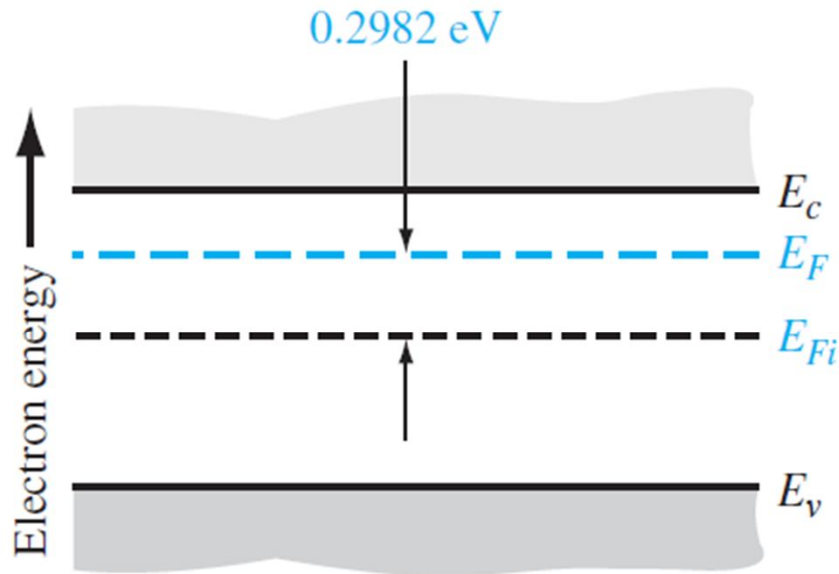
$$E_F - E_{Fi} = kT \ln \left(\frac{n_0}{n_i} \right) = 0.2982 \text{ eV}$$

In nonequilibrium, quasi Fermi level for electrons and holes becomes

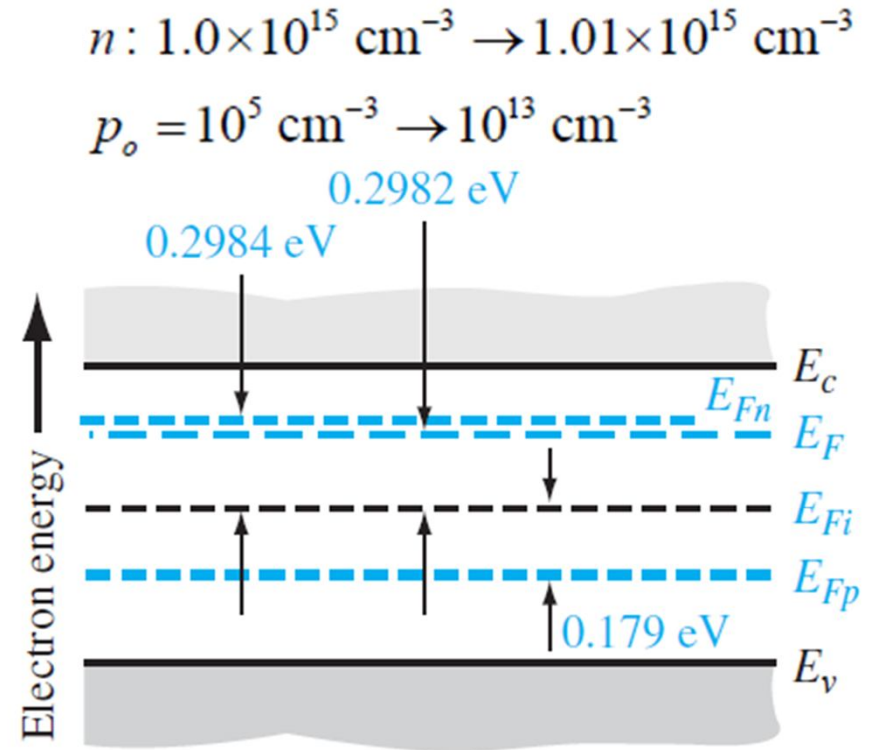
$$E_{Fn} - E_{Fi} = kT \ln \left(\frac{n_0 + \delta n}{n_i} \right) = 0.2984 \text{ eV}$$

$$E_{Fp} - E_{Fi} = kT \ln \left(\frac{p_0 + \delta p}{n_i} \right) = 0.179 \text{ eV}$$

QUASI-FERMI ENERGY LEVELS



(a)



(b)

- Since the majority carrier electron concentration does not change significantly for low-injection condition, the quasi-Fermi level for electrons here is not much different from the thermal-equilibrium Fermi level.
- The quasi-Fermi level for minority carrier holes is significantly the Fermi level and illustrate the fact that we have deviate from the thermal equilibrium significantly

Reading Materials

D. A. Neamen, “Semiconductor Physics and Devices: Basic Principles”, Fourth edition, Chapter 6, “Nonequilibrium Excess Carriers in Semiconductors”.